Spring Research Project: Mentor Rubric

Mentor Name(s): Poulomi Chakraborty

Mentee Name(s): Zane Johnson

Project Title: Stellar Models and the Chandrasekhar Limit

In a short paragraph, explain the project and its learning objectives:

The aim of the project is to introduce the student to the physical models that describe the structure and evolution of stars, with a focus on the **Chandrasekhar limit** and its implications for stellar evolution, especially for white dwarf stars. The learning objectives are described in the following:

- Understand basic stellar structure and the forces that govern stars.
- Learn how stars evolve, including how they form, burn fuel, and ultimately end their lives as white dwarfs, neutron stars, or black holes.
- Explore the **Chandrasekhar limit**, its significance, and its role in determining the fate of a star.
- Use basic physics (including equations of hydrostatic equilibrium and mass-luminosity relations) to model simple stellar properties.

What are the deliverables? (i.e. plot/figure/image/equation mentee will reach)

Mass- Luminosity relationship of stars. Plot using real data and verification. Chandrasekhar limit. Using real data, predicting fate of stars.

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Project Calendar: Indicate what the mentee will be expected to do each week. Work days are indicated in the left column. Include estimated time, resources, in/out of class work, and whether it is completed on their own or with you. The project should take 8-10 hours in total.

Dates:	Tasks:
Week 1 (2/10-2/16) Intro to Project	Learning prerequisites: Ideal gas law, Pressure (gas), Gravitational force, Radiation pressure
Week 2 (2/17-2/23) In Class Work Day	Learn about principle of radiative equilibrium and the luminosity of stars
Week 3 (2/24-3/2) In Class Work Day	Verify mass-luminosity relationship using actual star data. Optional: try out a calculation with different masses and see the competition between radiation pressure and gas pressure.
Week 4 (3/3-3/8) Class Lecture	Introduce dimensional analysis and scaling. Get the size and lifetime scales. Introduce mass deficit.
Spring Break (3/10-3/14)	None
Week 5 (3/17-3/23) In Class Work Day	Introduction to Nuclear Reactions that happen in stars.
Week 6 (3/24-3/30) Class Lecture	Using data (luminosity) available for 5 stars, and calculating their mass and radius and predicting their fate.
Week 7 (3/31-4/6) Class Lecture	Introducing Chandrasekhar limit and the fate of the stars.
Week 8 (4/7-4/13) In Class Work Day	Properties of white dwarves, neutron stars, black holes etc.
Week 9 (4/14-4/20) Poster Work Day	Make the poster.
Week 10 (4/21) Present!	

Reading List: (Make sure material is level appropriate.)

Giancoli – Physics for Scientists and Engineers for basic physics concepts
Any undergraduate for Astrophysics

Resource List: (This should include lab space, instruments, and software. Note that mentees are not allowed to complete research projects that require safety training operation of lab equipment. If you feel this is detrimental to your project, please petition Polaris leadership.)

Other Notes: